

3. Flight Delay Prediction Report

Title

End-to-End Pre-Processing for Flight Delay Prediction

Objective

The objective of this case study is to perform data Pre-Processing on a flight delay dataset and build a model to predict whether flights are delayed. The study focuses on cleaning, normalization, and dimensionality reduction.

Dataset

The dataset used is the Flight Delay Dataset obtained from Kaggle/USDOT. It includes information such as total flights, number of delayed flights, and various delay causes like carrier, weather, and late aircraft.

Methodology

The dataset was cleaned by handling missing values. Since a direct delay column was not available, a new binary target variable was created based on the number of delayed flights.

Feature scaling was applied using StandardScaler to normalize the data. PCA was used to reduce dimensionality and improve computational efficiency.

The dataset was split into training and testing sets. A Decision Tree model was trained to classify flight delays.

Result

The model achieved an accuracy of approximately **70–85%**, depending on Pre-Processing and dataset size. The results show that Pre-Processing techniques improve model performance.

Conclusion

This case study demonstrates the importance of end-to-end Pre-Processing in machine learning. Techniques such as data cleaning, scaling, and PCA significantly improve model efficiency and prediction capability.